

Xiaochuan Li

M.Sc. Human and AI · University of Technology Nuremberg

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RESEARCH INTERESTS

Formal and experimental semantics; psycholinguistics; language processing and learning; computational semantics; language-model representation, evaluation, and intervention. Current focus: epistemic modality and discourse alternatives.

EDUCATION

M.Sc., Human and Artificial Intelligence

2025 – 2027

University of Technology Nuremberg (UTN)

Expected graduation: Summer 2027. Coursework: cognitive psychology, experimental design and statistics, interpretability, deep learning, epistemology.

Student · Preparatory Studies

2024 – 2025

University of Marburg · enrolled student; intensive German for graduate study

Comparing German with Chinese and English made grammatical differences—case, word order, and clause structure—concrete rather than abstract.

B.A., Philosophy

2020 – 2024

Shenzhen University

Senior thesis on Leibniz's *characteristica universalis* (advisor: Zang Yong). Coursework in philosophy of language, analytic philosophy, logic, and mathematical logic.

CURRENT RESEARCH: MODALITY

Three Ways of Leaving p Unsettled — analysis in development

A formal semantic and pragmatic comparison of might p, might p or might not p, and I don't know whether p, examining the commitments they create and the responses they license.

M.Sc. thesis: Modal language in preschool children — in planning

Planned alongside an existing study of how preschool children reason about possibilities; design being agreed, no data yet.

Modal expressions in language models — in development

Diagnostics on pretrained models; controlled training of small models is planned.

PAPERS AND MANUSCRIPTS

Typicality in referent choice: what the description leaves open — working paper

Superordinate descriptions, typicality, and what forced-choice responses can show. [LingBuzz](#) · [DOI](#)

Prompt framing and latent control states — manuscript in preparation

Designed controlled dilemma comparisons; experimental analysis of residual-stream interventions, low-rank effects, and competing explanations. Ongoing collaboration with DFKI.

COURSE AND TECHNICAL PROJECTS

Cross-lingual ProtoBias — single-author course project

Controlled evaluation across seven languages and two VLM families; approximately 12,600 judgments, item-clustered analysis, human controls, and pipeline-integrity checks. [Report](#) · [Code](#)

Street-view country classification — team course project

Trained a CNN from random initialization and ran controlled experiments for 18-country classification. My contribution: training, ablations, and performance analysis; a teammate led the core implementation.

METHODS AND TECHNICAL SKILLS

Programming and data. Python, NumPy, pandas, scikit-learn, Git, LaTeX.

ML / NLP. PyTorch, Hugging Face Transformers, model inference, dataset pipelines, LoRA fine-tuning.

Interpretability. Linear and structural probing, activation patching, steering, ablation (working proficiency).

Experimental design and statistics. Controlled stimuli, minimal pairs, preregistration, mixed-effects analysis (working proficiency).

LANGUAGES

Mandarin Chinese (native), German (C1), English (C1), Japanese (learning).